

Vertex cover pesato e non

Grafo $G = (V, E)$

Tramite solver:

Modello PLI vertex cover non pesato:

$$\min \sum_{u \in V} x_u$$

$$x_u + x_v \geq 1 \quad \forall (u,v) \in E$$

$$x_u \in \{0,1\} \quad \forall u \in V$$

Modello PLI vertex cover pesato:

$$\min \sum_{u \in V} c_u x_u$$

$$x_u + x_v \geq 1 \quad \forall (u,v) \in E$$

$$x_u \in \{0,1\} \quad \forall u \in V$$

Algoritmo vertex cover non pesato 1 (partendo da un vertice qualunque):

1. $C = \{\}$; #cover
2. $I = E$; #archi "scoperti"
3. for v in V {
4. $C.insert(v)$;
5. $I = I \setminus \delta(v)$;
6. if ($I.isEmpty$) then return C ;
7. }

Algoritmo vertex cover pesato 1 (adattato da me, partendo dal vertice di peso minimo):

1. $C = \{\}$; #cover
2. $I = E$; #archi "scoperti"
3. $W = \text{sort_by_weight}(V)$; #vertici ordinati per peso crescente
4. for v in W {
5. $C.insert(v)$;
6. $I = I \setminus \delta(v)$;
7. if ($I.isEmpty$) then return C ;
8. }

Algoritmo vertex cover non pesato 2 (ordinando i vertici per grado decrescente):

```
1. C = {}; #cover
2. I = E; #archi "scoperti"
3. W = sort_by_deg(V); #vertici ordinati per grado decrescente
3. for v in W {
4. C.insert(v);
5. I = I /  $\delta(v)$ ;
6. if (I.isEmpty) then return C;
7. }
```

Algoritmo vertex cover pesato 2 (adattato da me, ordinando i vertici per grado e peso):

```
1. C = {}; #cover
2. I = E; #archi "scoperti"
3. W = sort_by_deg_and_weight(V); #vertici ordinati per grado decrescente
3. for v in W {
4. C.insert(v);
5. I = I /  $\delta(v)$ ;
6. if (I.isEmpty) then return C;
7. }
```

Algoritmo vertex cover non pesato 3 (matching massimale):

```
1. C = {}; #cover
2. I = E; #archi "scoperti"
3. while( ! I.isEmpty){
4. e = I[0]; #prendo un arco scoperto
5. u = first_vertex(e);
6. v = second_vertex(e);
7. C.insert(u);
8. C.insert(v);
9. J = e  $\cup \delta(u) \cup \delta(v)$ ;
10. I = I / J;
11. }
12. return C;
```

Algoritmo vertex cover pesato 3 (adattato da me, matching massimale):

1. $C = \{\}$; #cover
2. $I = E$; #archi "scoperti"
3. while(! I.isEmpty){
4. $W = \text{sort_by_weight}(I)$; #ordino gli $e = (u,v) \in E$ in ordine crescente di $c_u + c_v$
5. $e = W[0]$; #prendo il minimo arco scoperto
6. $u = \text{first_vertex}(e)$;
7. $v = \text{second_vertex}(e)$;
8. $C.\text{insert}(u)$;
9. $C.\text{insert}(v)$;
10. $J = e \cup \delta(u) \cup \delta(v)$;
11. $I = I / J$;
12. }
13. return C;

Algoritmo vertex cover non pesato 4 (doubling and rounding down):

1. *Risolve il problema rilassato:*

$$\min \sum_{u \in V} x_u$$

$$x_u + x_v \geq 1 \quad \forall (u,v) \in E$$

$$x_u \geq 0 \quad \forall u \in V$$

2. $w = 0^{|V|}$;
3. for ($i \leftarrow 1$ to $|V|$) {
4. if($x_i^* \geq \frac{1}{2}$) then $w_i = 1$;
5. }
6. return w

Algoritmo vertex cover pesato 4 (adattato da me, doubling and rounding down):

1. *Risolve il problema rilassato:*

$$\min \sum_{u \in V} c_u x_u$$

$$x_u + x_v \geq 1 \quad \forall (u,v) \in E$$

$$x_u \geq 0 \quad \forall u \in V$$

2. $w = 0^{|V|}$;
3. for ($i \leftarrow 1$ to $|V|$) {
4. if($x_i^* \geq \frac{1}{2}$) then $w_i = 1$;

```

5. }
6. return w

```

Algoritmo vertex cover non pesato (basato sul duale, adattato da me impostando tutti i pesi a 1):

```

1.  $x = 0^{|V|}$ ;
2.  $y = |0|^{|E|}$ ;
3.  $I = E$ ; #archi "scoperti"
4. while( $x \neq \text{vertex\_cover}(G)$ ){
5.  $e = I[0]$ ; #prendo un arco scoperto
6.  $u = \text{first\_vertex}(e)$ ;
7.  $v = \text{second\_vertex}(e)$ ;
8.  $\text{increment\_u} = 1 - \sum_{e \in \delta(u)} y_e$ ;
9.  $\text{increment\_v} = 1 - \sum_{e \in \delta(v)} y_e$ ;
10. if( $\text{increment\_u} < \text{increment\_v}$ ) then {
11.  $y_e = 0$ ;
12.  $x_u = 1$ ;
13.  $I = I \setminus \delta(u)$ ;
14. } else if( $\text{increment\_u} > \text{increment\_v}$ ){
15.  $y_e = 0$ ;
16.  $x_v = 1$ ;
17.  $I = I \setminus \delta(v)$ ;
18. } else {
19. #posso scegliere uno dei due
20.  $y_e = \text{increment\_u}$ ; #poiché sono uguali
21. if( $\deg(u) \leq \deg(v)$ ){ # scelgo il vertice con il grado maggiore
22.  $x_v = 1$ ;
23.  $I = I \setminus \delta(v)$ ;
24. } else {
25.  $x_u = 1$ ;
26.  $I = I \setminus \delta(u)$ ;
27. }
28. }
29. }
30. return x;

```

Algoritmo vertex cover pesato (basato sul “duale”):

```
1.  $x = 0^{|V|}$ ;  
2.  $y = |0|^{|E|}$ ;  
3.  $I = E$ ; #archi “scoperti”  
4. while( $x \neq \text{vertex\_cover}(G)$ ){  
5.  $e = I[0]$ ; #prendo un arco scoperto  
6.  $u = \text{first\_vertex}(e)$ ;  
7.  $v = \text{second\_vertex}(e)$ ;  
8.  $\text{increment\_}u = c_u - \sum_{e \in \delta(u)} y_e$ ;  
9.  $\text{increment\_}v = c_v - \sum_{e \in \delta(v)} y_e$ ;  
10. if( $\text{increment\_}u \leq \text{increment\_}v$ ) then {  
11.  $y_e = \text{increment\_}u$ ;  
12.  $x_u = 1$ ;  
13.  $I = I / \delta(u)$ ;  
14. } else {  
15.  $y_e = \text{increment\_}v$ ;  
16.  $x_v = 1$ ;  
17.  $I = I / \delta(v)$ ;  
18. }  
19. }  
20. return  $x$ ;
```

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